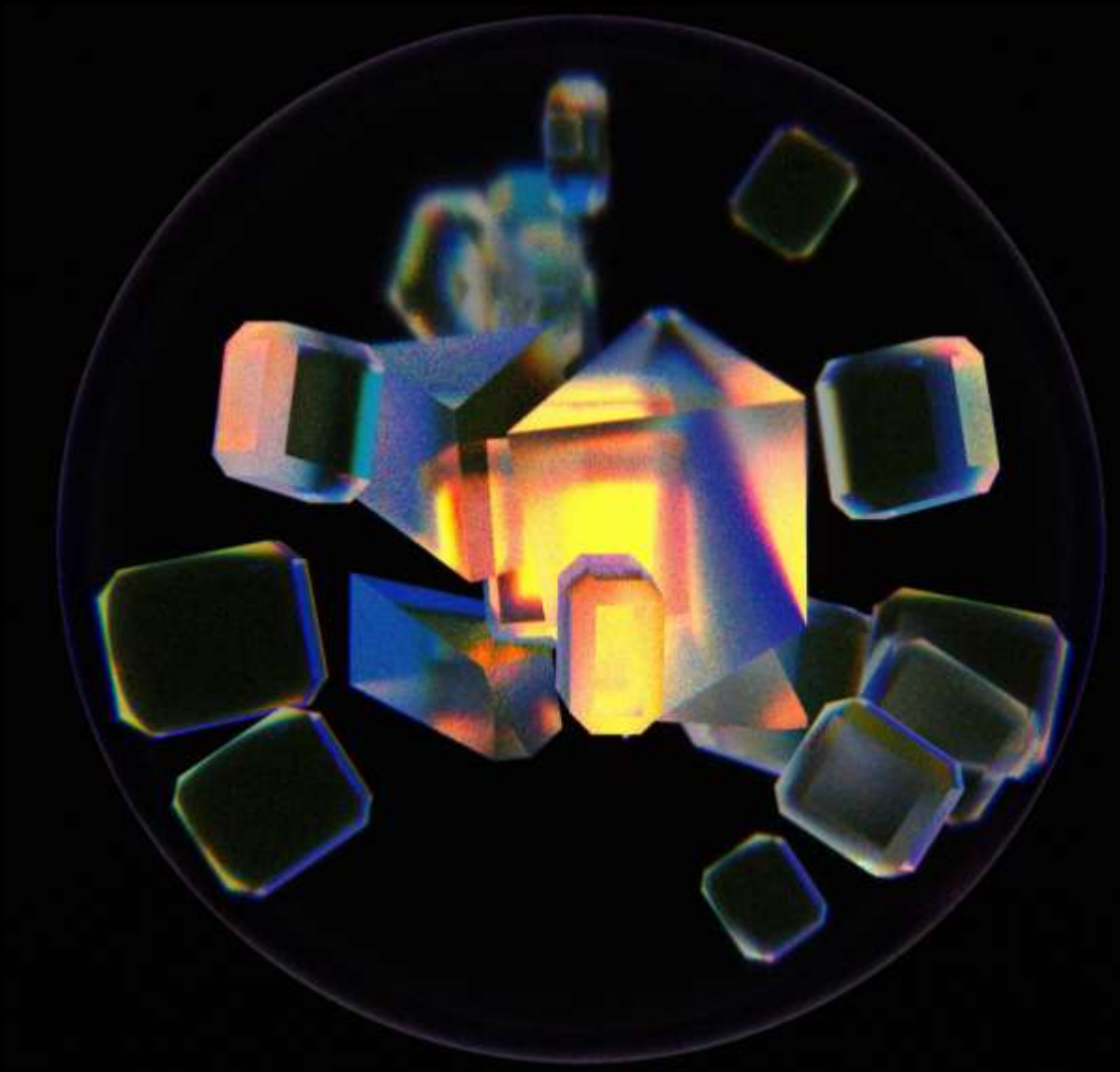
UNIVERSITY OF TORONTO | Acceleration Consortium

CANADA FIRST RESEARCH EXCELLENCE FUND | APOGÉE CANADA FONDS D'EXCELLENCE EN RECHERCHE | UBC THE UNIVERSITY OF BRITISH COLUMBIA

Owen A. Melville, PhD
Staff Scientist, Polymer SDL, Acceleration Consortium

Resource-Efficient Bayesian Optimization for Self-Calibrating Liquid Handling

2026 MRS Spring Meeting & Exhibit
April 26, 2026



Overview

1. Polymer Self-Driving Lab @ Acceleration Consortium
2. Why focus on liquid handling?
3. Why use Bayesian optimization for liquid handling?
4. How to use Bayesian optimization for liquid handling
5. Live calibration demo with audience interaction
6. Discussion of details

1. Polymer Self-Driving Lab

Polymer Self-Driving Lab – Acceleration Consortium

Staff Scientists

Chanelle Brown

Nipun Gupta

Owen Melville

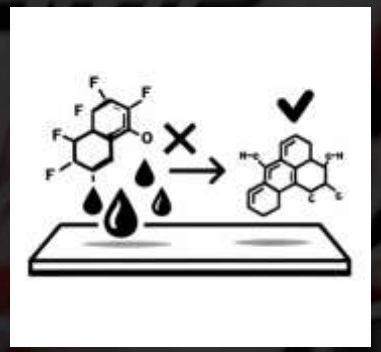
Harrison Mills

Faculty Advisors

Prof. Helen Tran

Prof. Jay Werber

Application Areas



PFAS Replacement

AI-Guided Synthesis

Surfactants

Flexible Automation

OUR LAB WILL BE BROAD, FLEXIBLE, AND MODULAR

Polymer Synthesis

Flow reactor (Vaportec)
GPC (Waters/Wyatt)

- Automated flow synthesis, sample fractionation and vial capping
- Automated purification under development
- Reaction monitoring using automated GPC + MALS/RID

Mechanical Characterization

Liquid Handler (OpenTrons)
Arm (Universal Robotics)
Indenter (Custom)

- Combine resins & viscous polymer precursors
- Spot UV curing
- Stiffness, creep, and relaxation properties analysis via indentation
- Identify formulations with target mechanical properties

Optical Characterization

Plate Reader (Agilent)
Robotic Track + Station (North Robotics)

- Quickly create complex polymer workflows
- Use AI-guided calibration
- Measure responses from optical probes or dyes
- Run workflows up to 72h

Flexible Automation

- Create SDLs with CNC machines:
- *Custom photoreactor*
- *Budget liquid dispenser*
- *Fraction collector*
- *Contact angle measurement*
- And more!

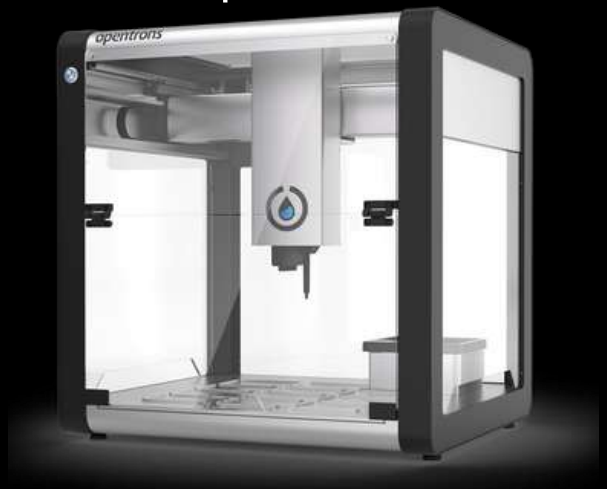
2. Why Focus on Liquid Handling?

Many Automated Systems for Liquid Handling

Hamilton

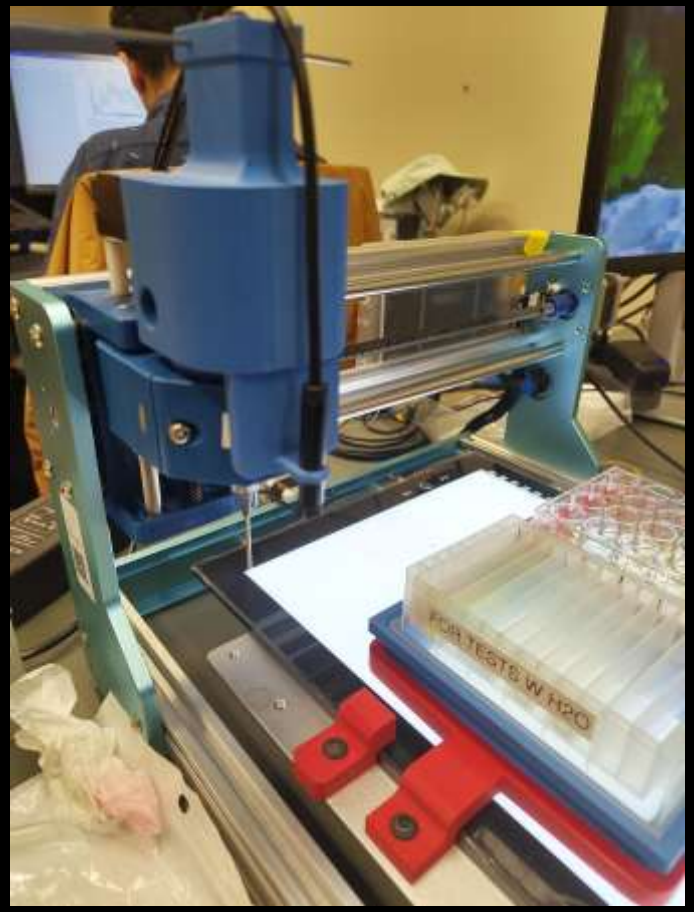
https://www.hamiltoncompany.com/knowledge-base/brochure/microlab-star-brochure

OpenTrons



https://opentrons.com/products/ot-2-robot?

North Robotics N9
(Air displacement from syringe pump)



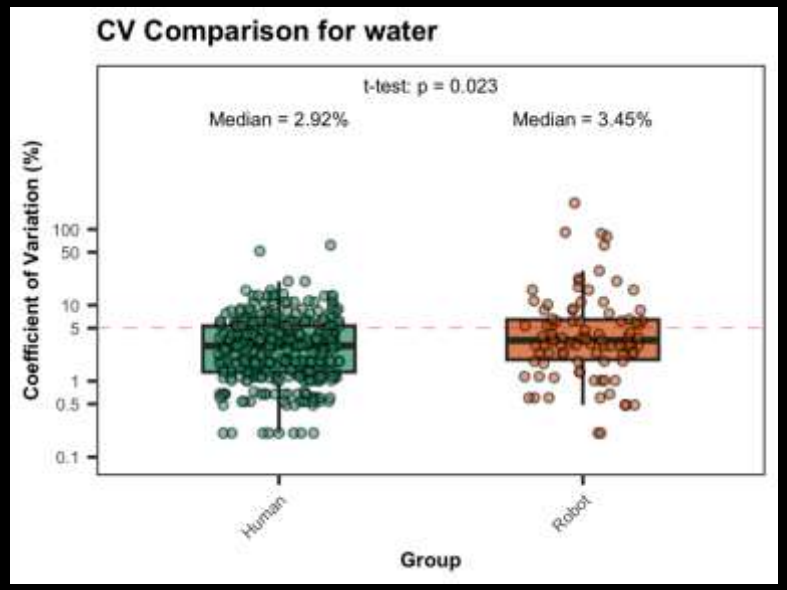
Budget CNC Dispenser
(Actuator + Syringe)

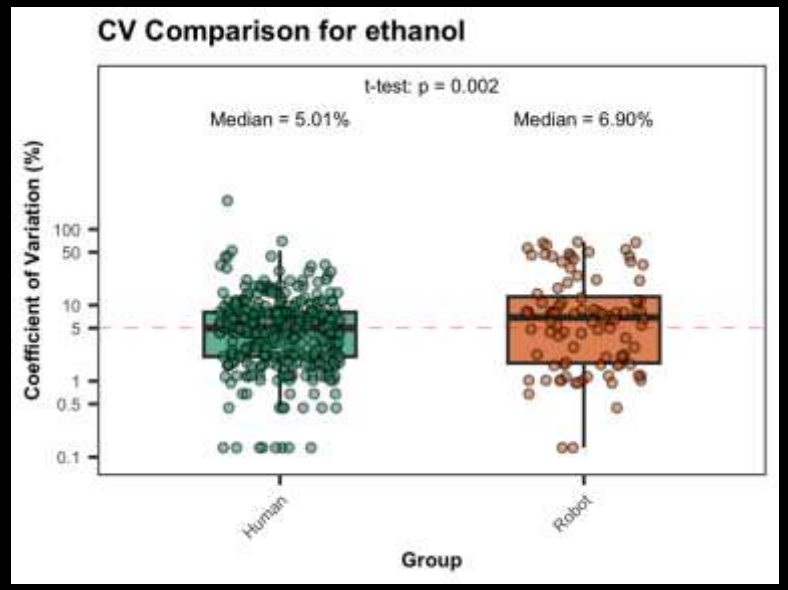
Different Liquid = Different Difficulty

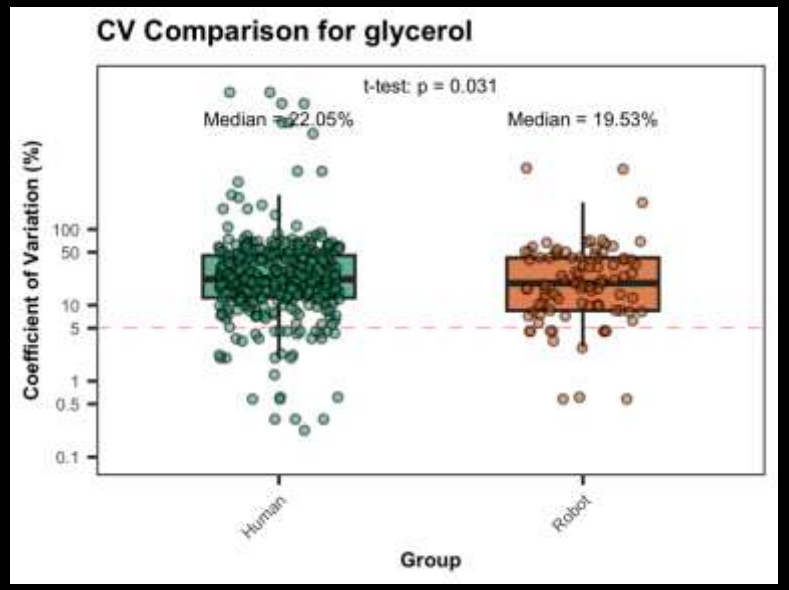
Ilya Yakavets

Wei Yang Tham

- **Question**: Do robots pipet better than humans?
- **Answer**: Depends on liquid
- Glycerol much harder to pipet
  - Even robots with liquid classes fared poorly
  - *I was a human volunteer and personally struggled*

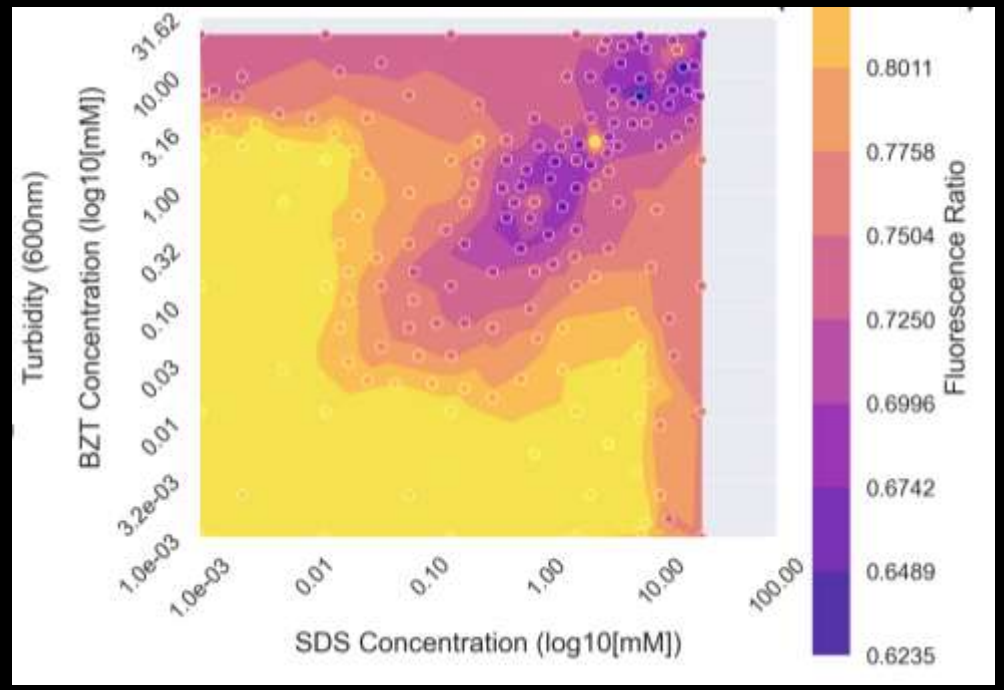


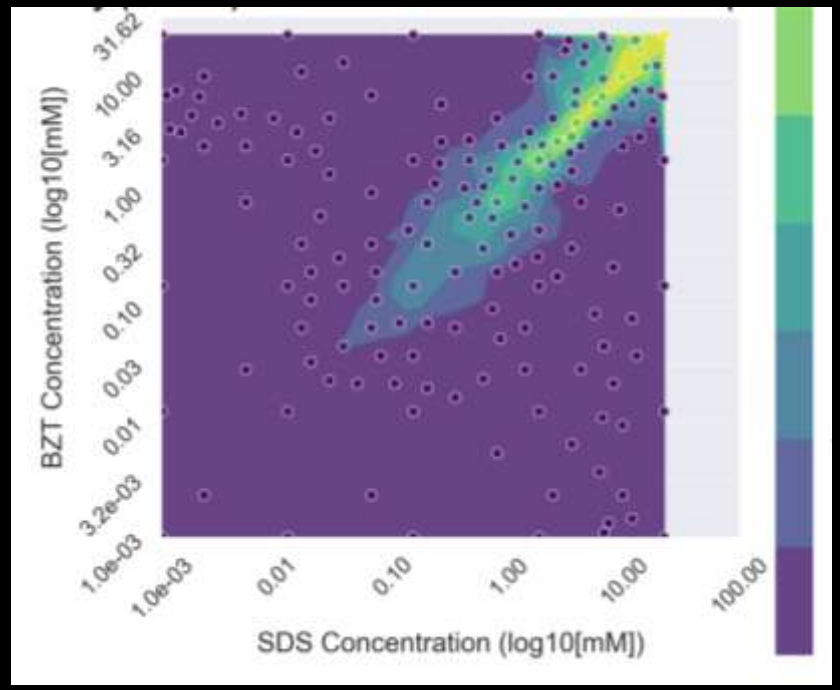




Liquid Handling is Important for Self-Driving Labs

- **Example Below**: 733 transfers in one 2D surfactant boundary mapping workflow, 3 liquid types
- **Concentrations** are often inputs for active learning models
- **Issues with pipetting** can compound systematic uncertainty, reduce reproducibility and disrupt an active learning model
- **People need to trust their data, especially when using automation + AI**

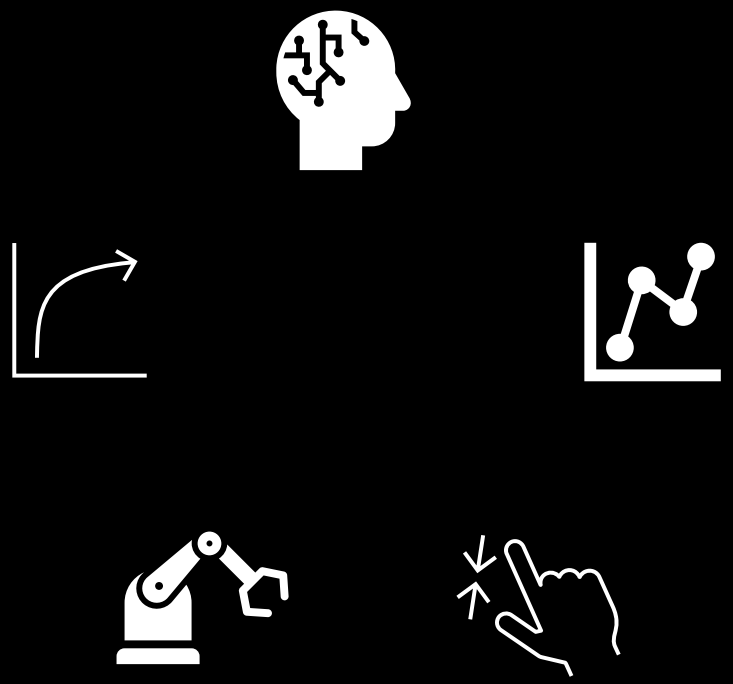




3. Why use Bayesian Optimization for Liquid Handling?

Many Options for Monitoring Liquid Handling

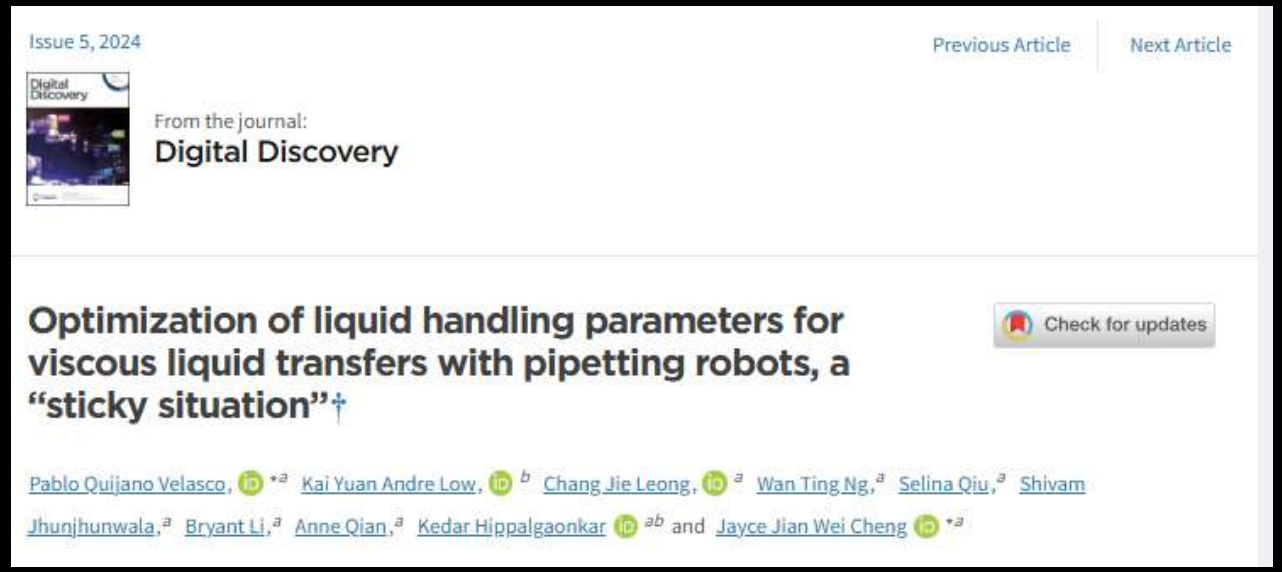
- Use default liquid classes from supplier
- Tune parameters by hand (DIY approach)
- Use feedback-based monitoring
- Use design of experiments (DOE)
- Use an active learning algorithm
  - **** Bayesian optimization ****
  - Large language model (LLM)



Goal: Find a way to optimize pipetting parameters
(Aspirate speed, blowout volume… etc.)
so that we can dispense accurately & quickly

Why is Bayesian Optimization Great?

- Efficient use of measurements
- Requires no physics
  (makes its own model of data)
- Compatible with noisy data
- Can work with multiple objectives
  (eg time and accuracy)
- Balances exploration and exploitation
- Frameworks available online


Issue 5, 2024

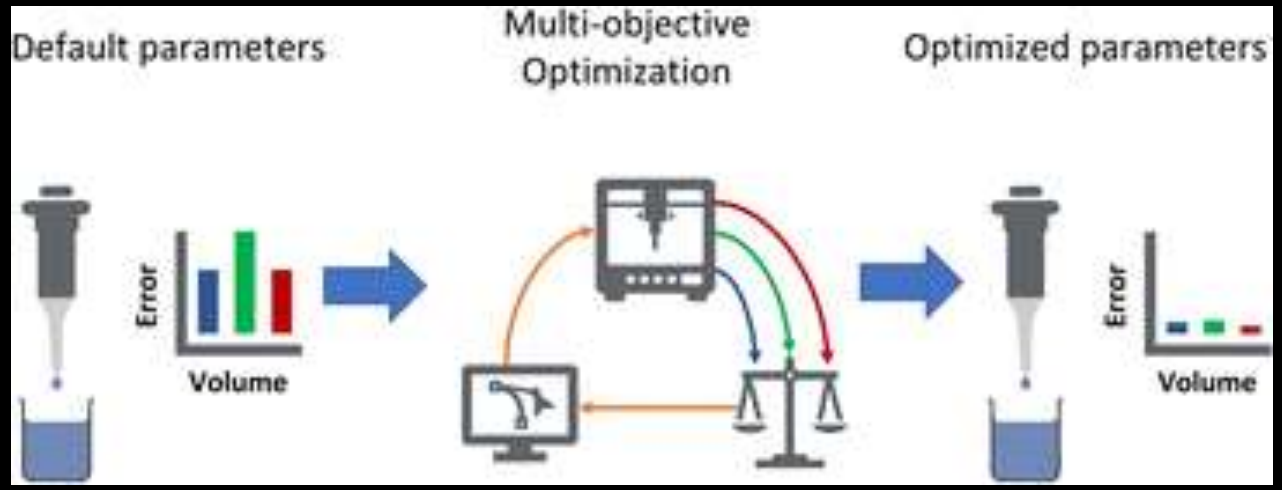
Previous Article Next Article

From the journal:
Digital Discovery

Optimization of liquid handling parameters for viscous liquid transfers with pipetting robots, a "sticky situation"†

Check for updates

Pablo Quijano Velasco,[a] Kai Yuan Andre Low,[b] Chang Jie Leong,[a] Wan Ting Ng,[a] Selina Qiu,[a] Shivam Jhunjhunwala,[a] Bryant Li,[a] Anne Qian,[a] Kedar Hippalgaonkar[ab] and Jayce Jian Wei Cheng[a]



4. How to Use BO for Liquid Handling

BO Implementation: Step 1

1. Define optimizer
   - Select Bayesian Optimization framework: **Ax**, Baybe, Botorch, etc.
   - Depending on framework: Acquisition function, generation strategy, etc.

```
# Step 1: Create generation strategy (pure Bayesian)
generation_strategy = GenerationStrategy(
    steps=[
        # Pure Bayesian optimization from the start
        GenerationStep(
            model=Models.GPEI,              # Gaussian Process + Expected Improvement
            num_trials=-1,                   # Continue indefinitely
        ),
    ]
)

# Step 2: Create Ax client with custom strategy
ax_client = AxClient(generation_strategy=generation_strategy)
```

BO Implementation: Step 2

2. Define parameters and bounds
   - Aspirate & dispense speed
   - (Post) Aspirate and dispense wait time
   - Pre & post aspirate air volumes (not used today)
   - Blowout volume (syringe pump)
   - Over-aspirate volume (extra liquid)

```
parameters = [
    # Speed parameters
    {"name": "aspirate_speed", "type": "range", "bounds": [2.0, 30.0], "parameter_type": "float"},
    {"name": "dispense_speed", "type": "range", "bounds": [2.0, 30.0], "parameter_type": "float"},

    # Volume parameters
    {"name": "overaspirate_vol", "type": "range", "bounds": [0.001, 0.050], "parameter_type": "float"},
    {"name": "blowout_vol", "type": "range", "bounds": [0.000, 1.000], "parameter_type": "float"},

    # Timing parameters
    {"name": "aspirate_wait_time", "type": "range", "bounds": [0.0, 10.0], "parameter_type": "float"},
    {"name": "dispense_wait_time", "type": "range", "bounds": [0.0, 10.0], "parameter_type": "float"},
]
```

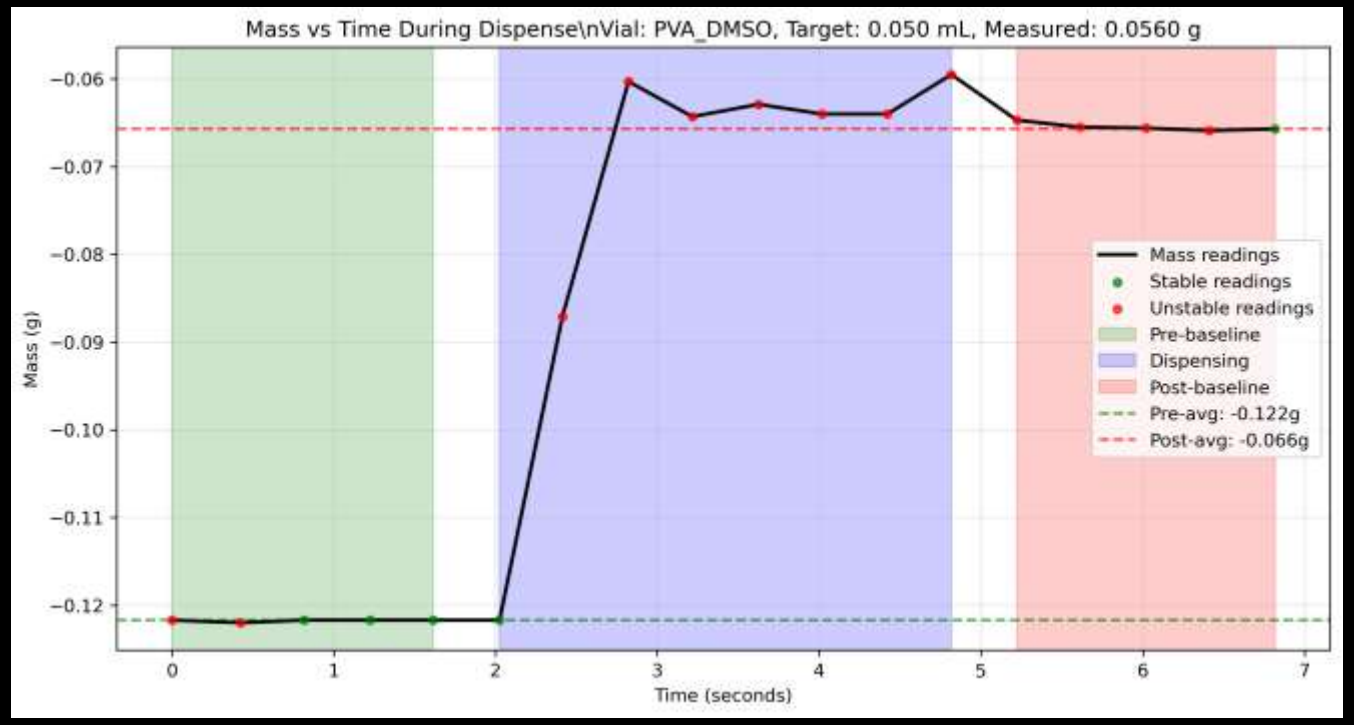
BO Implementation: Step 3

3. Decide what outputs to optimize
   - Accuracy: Minimize average deviation (%)
   - Precision: Minimize replicate CV (variability) (%)
   - Time: Minimize elapsed time (s)

```
objectives = {
    "deviation": ObjectiveProperties(minimize=True, threshold=50.0),
    "variability": ObjectiveProperties(minimize=True, threshold=10.0),
    "time": ObjectiveProperties(minimize=True, threshold=90.0),
}
```

BO Implementation: Step 4

4. Have a method to measure outputs
   - Measure mass & convert to volume with density (see below) -> accuracy
   - Use a timer for time (simple in python)
   - Use replicates for precision (we will use 2)



BO Implementation: Step 5

5. Initialize the optimizer
   - Optimizer needs seed data to model and pick from
     - **Option 1**: Use previous experimental measurements
     - **Option 2**: Use manual measurements (What we will do today)
     - **Option 3**: Generate seed points using space-filling designs (e.g. Sobol, Latin hypercube) and measure responses

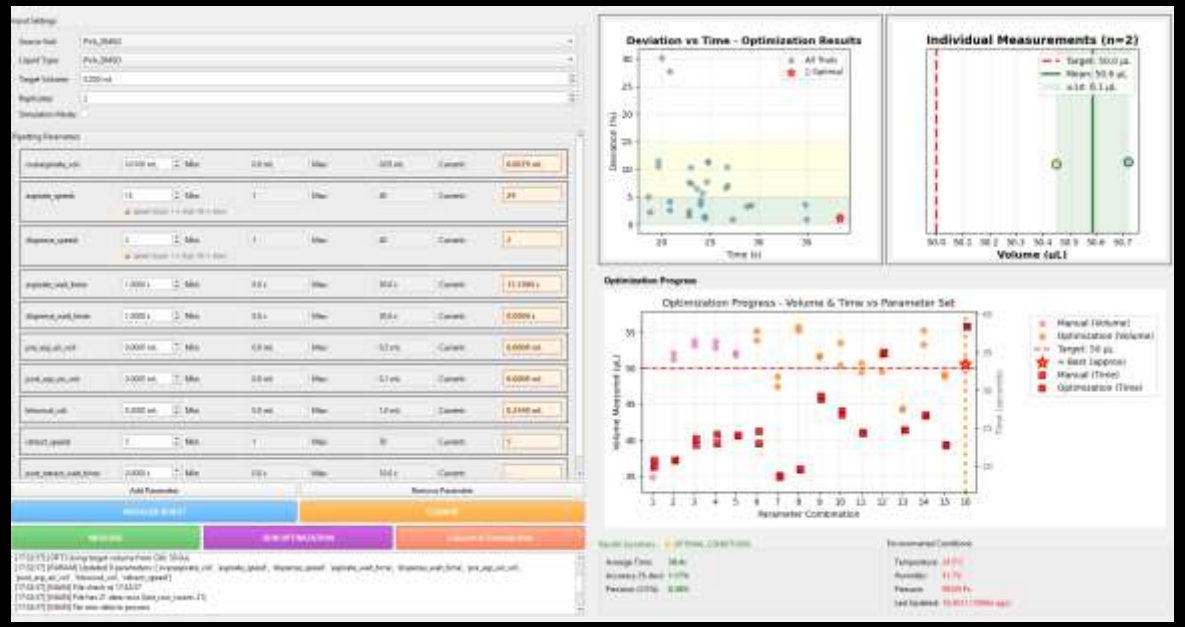
```python
ax_client.create_experiment(
    name="pipetting_optimization",
    parameters=parameters,
    objectives=objectives,
)
```

```python
# Seed with historical experimental data
for result in previous_experiments:
    parameters = {
        "aspirate_speed": result.aspirate_speed,
        "dispense_speed": result.dispense_speed,
        "overaspirate_vol": result.overaspirate_vol,
        # ... other parameters
    }

    # Attach the trial to Ax
    trial_index = ax_client.attach_trial(parameters=parameters)

    # Complete with measured results
    ax_client.complete_trial(
        trial_index=trial_index,
        raw_data={
            "deviation": result.accuracy_error,
            "variability": result.precision_error,
            "time": result.cycle_time,
        }
    )
```

5. Live Liquid Handling DEMO

Live Demo

- North Robotics system (now enclosed)
- View system via livestream (if lights are left on)
- Run Demo GUI using Remote Desktop through Tailscale
  1. Manually guess some parameter combinations and measure outcomes
  2. Run optimization
- Liquid: 6 wt% PVA (~10 kDa) in DMSO
- System will aspirate and dispense into the same vial using parameters, switching tips each time

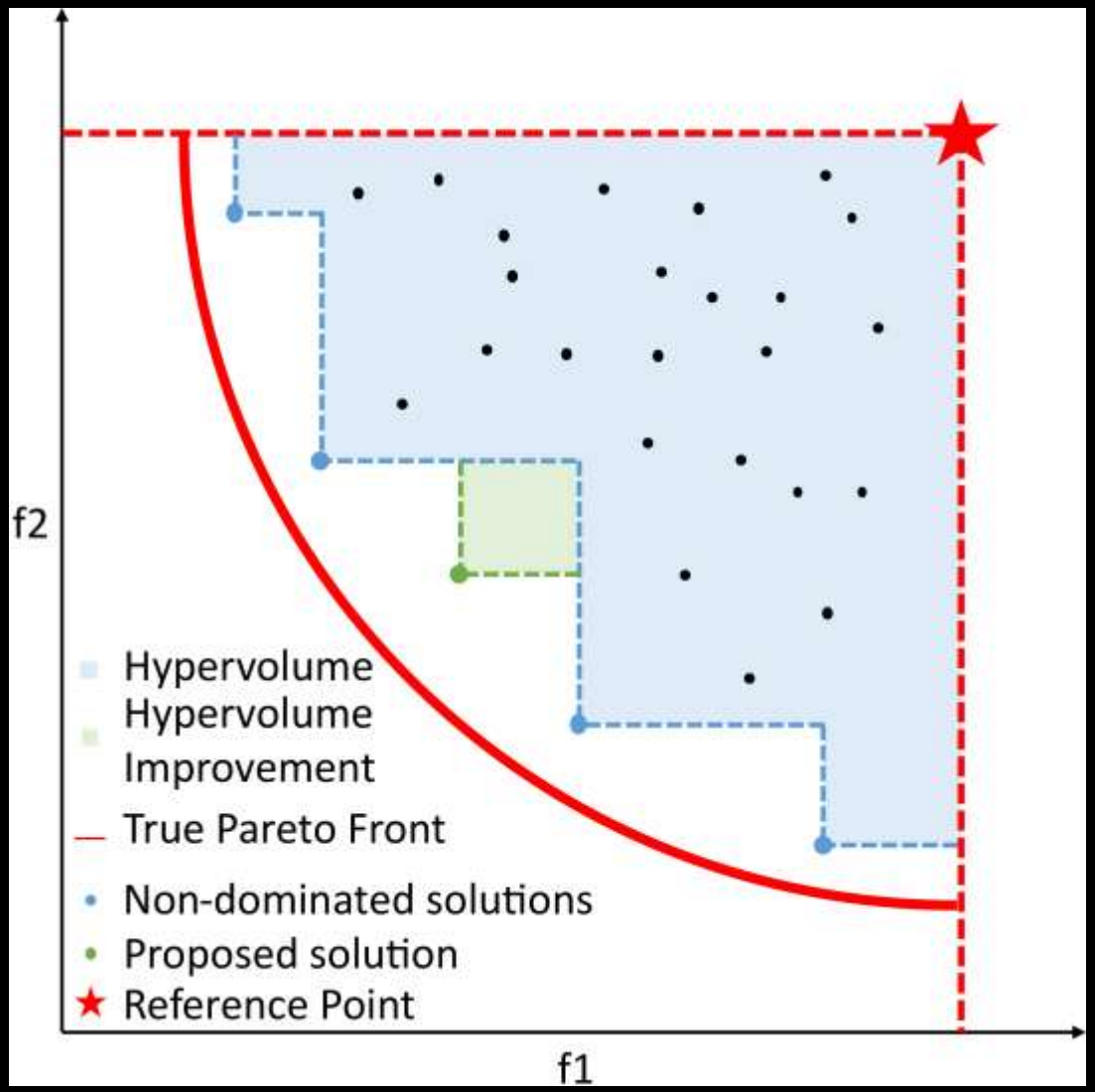


6. Details & Discussion

All about Outputs

- Single-objective optimization
- Multi-objective (MO) optimization
  - Weighted-sum approach $f = \sum w_i y_i$
  - Pareto-front approach (hypervolume)
  - Typically need normalization (handled under the hood by Ax)

- **My experiments**: inconclusive which MO approach is best for calibration



Low, A. K. Y.; Vissol-Gaudin, E.; Lim, Y. F.; Hippalgaonkar, K. Mapping pareto fronts for efficient multi-objective materials discovery. *J. Mater. Inf.* **2023**, *3*, 11. http://dx.doi.org/10.20517/jmi.2023.02

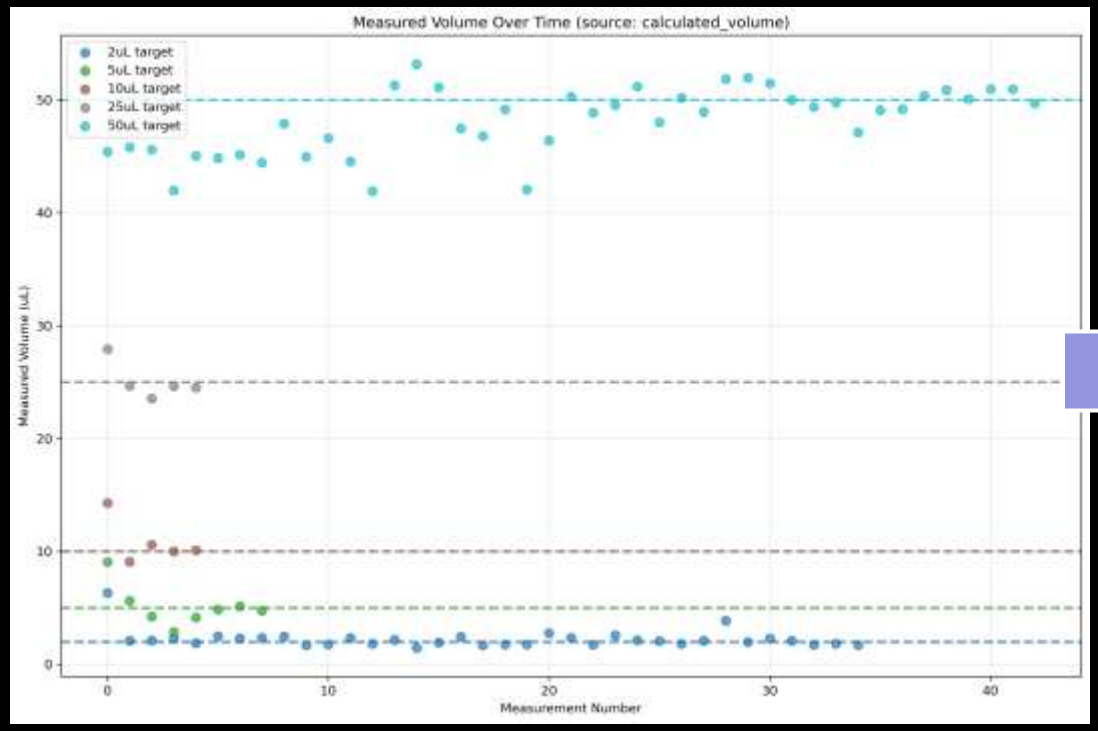
Moving from Optimization to Calibration Tool

- Stopping criteria
  - Experimental budget reached → 96x 200 uL tips
  - Enough suitable candidates found → Define tolerances that define suitability
- Evaluate/rank final candidates → Define mathematical formula to balance time, accuracy and precision
- Quickly tune other volumes → Adjust overaspirate volume, leave other parameters unchanged

My System

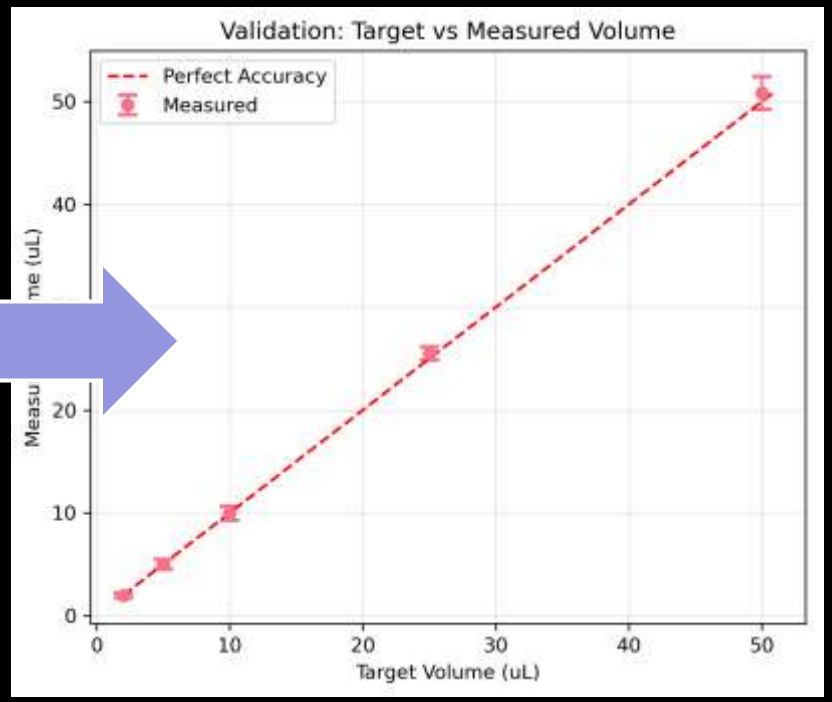
Previous Results – 2% Alginate Solution

Training



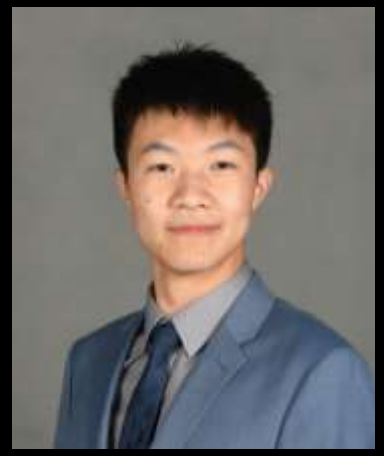
Best Conditions

Validation



- North robotics system
- Framework: Ax
- Acquisition: Expected improvement (Models.GPEI)

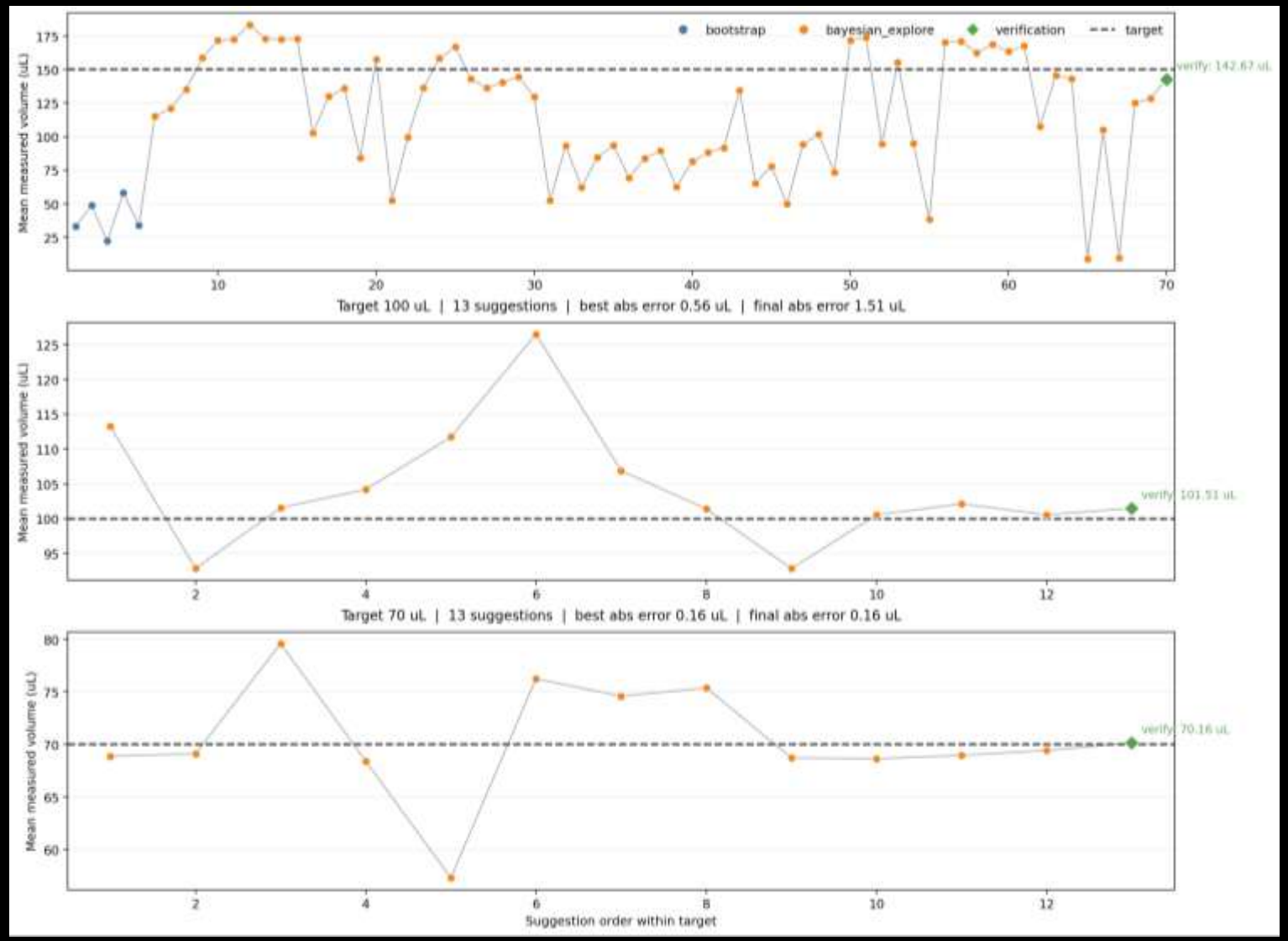
Previous Results – Glycerol



Edison Lin

Details:

- Hamilton liquid handler
- Framework: Ax
- Acquisition: Hypervolume (qNEHVI)



Can we use an LLM instead?

Convert Existing Data to JSON

↓

Create Structured Prompt

↓

Send to Remote LLM

↓

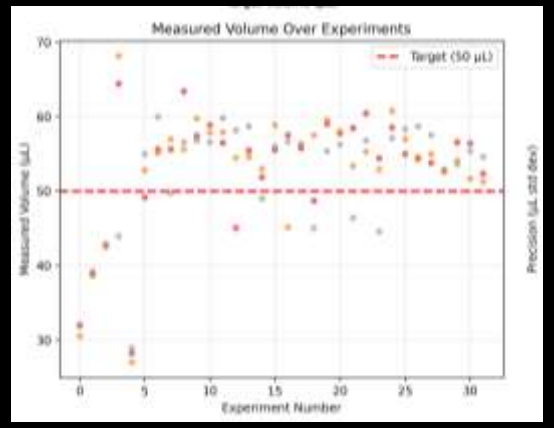
Parse Response

↓

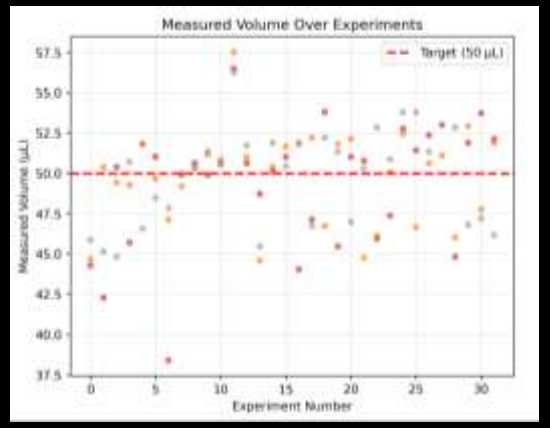
Test New Conditions

Glycerol dispense optimized with LLM

Starting Fresh



With Data



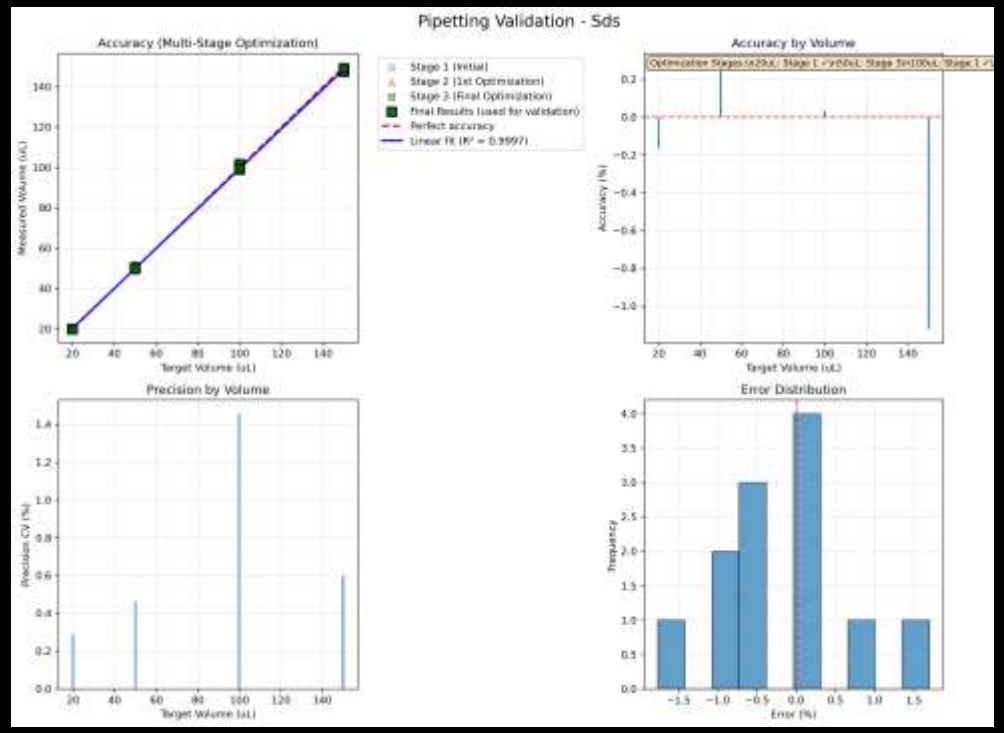
Advantages:

1. Use physics-based or explicit knowledge

Disadvantages:

1. Harder to setup
2. No built-in holistic data analysis
3. Need a hosted LLM or LLM service

Monitoring/Validation Builds Trust

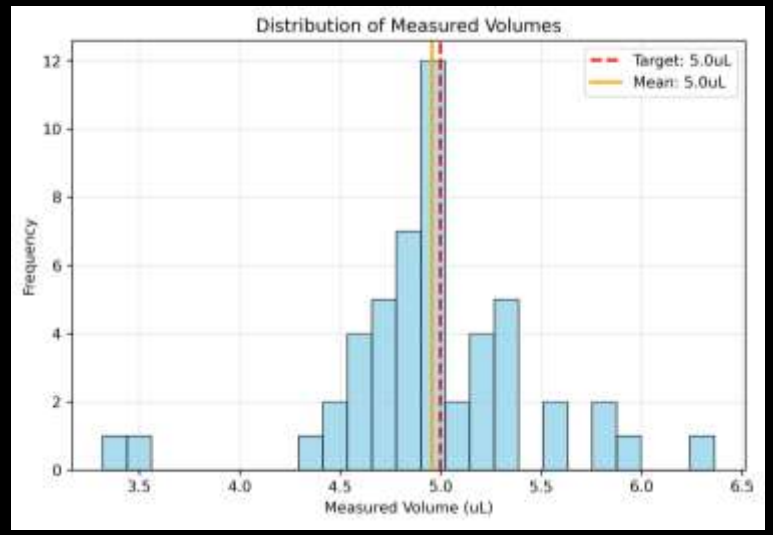


Found from Monitoring:

1. My DMSO-pyrene mixture at 5 uL was occasionally not pipetting at all after many dispenses (see below)
2. Smaller tips do not hold always hold their calibrations over time
3. Vial accidentally empty!

Pre-Workflow Dispense Validation:

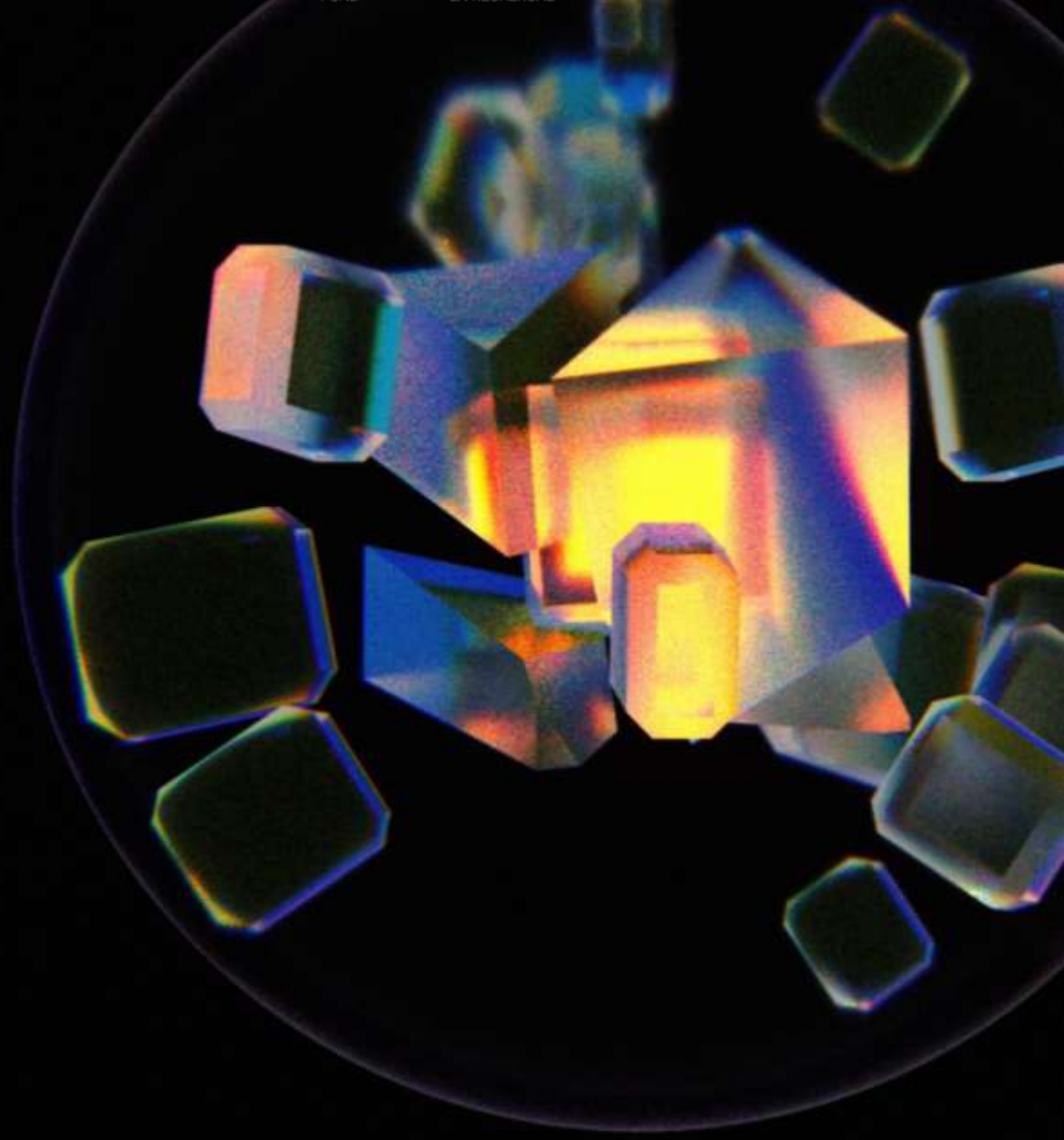
1. Do for each liquid (here SDS in water) for the ranges used in the workflow
2. Ensure everything looks right
3. Micro-adjustments if needed →
4. Record humidity, temperature etc.



UNIVERSITY OF TORONTO | Acceleration Consortium

CANADA FIRST RESEARCH EXCELLENCE FUND | APOGÉE CANADA FONDS D'EXCELLENCE EN RECHERCHE | UBC THE UNIVERSITY OF BRITISH COLUMBIA

Thank you!



EMAIL
owen.melville@utoronto.ca

WEBSITE
ACCELERATION.UTORONTO.CA

2026 MRS SPRING MEETING & EXHIBIT

https://www.surveymonkey.com/r/6MGMBTH

Thank you for completing the 2026 MRS Spring Meeting tutorial evaluation!